PROJECT: AUTOMATING EMPLOYEE DATA

TEAM MEMBERS

Team leader: AHAMED THOUFIK.T

Team members: ARAVIND R

Team members: CHANDRAN R

Team members: DINESH N

Team member: HARSHATH S

PROJECT REPORT

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1.INTRODUCTION

1.1 PROJECT OVERVIEW:

In modern organizations, employee data is continuously generated from multiple sources such as recruitment portals (ATS), HR spreadsheets, and biometric systems. Managing this personnel data manually is often error-prone and time-consuming, especially when linking employees to their respective departments, managers, and locations.

This project focuses on automating the employee data import process using Import Sets and Dot Walking. Import Sets allow employee records to be staged and validated before being pushed to the live User (sys_user) table. Transform Maps ensure that fields like 'Email', 'Employee ID', and 'Job Title' are accurately mapped. Dot Walking is utilized to seamlessly link employees to their managers’ details and department heads across different tables, ensuring a consistent and updated organizational hierarchy without manual data entry.

1.2 PURPOSE OF THE PROJECT

The primary goal is to simplify and automate the employee lifecycle data management. Manual updates often lead to "ghost employees," duplicate profiles, or incorrect reporting lines. This project aims to:

Ensure Data Integrity: Automatically validate HR data before it enters the system.

Maintain Relationships: Use Dot Walking to keep manager-subordinate relationships accurate across the platform.

Increase Efficiency: Reduce the time taken for bulk employee onboarding and offboarding.

Minimize Errors: Eliminate human mistakes in sensitive data like salary grades or department codes.

2.IDEATION PHASE

2.1 BRAINSTORMING & IDEA PRIORITIZATION

Step 1 – Brainstormed Ideas

Manual entry of employee records into the User table.

Script-based imports (Fix Scripts/Background Scripts) for bulk hiring.

Standard Import Sets for basic employee field mapping.

Automated Import Sets with Transform Maps and Dot Walking for complex organizational hierarchy.

Step 2 – Grouping

Manual Solutions:

One-by-one record creation (Slow and error-prone).

Script-based Solutions:

Using Excel imports with heavy script logic (High maintenance).

Semi-Automated Solutions:

Basic Import Sets without relationship mapping (Requires manual manager linking).

Fully Automated Solutions (Selected):

Automated Import Sets with Transform Maps for data cleansing.

Dot Walking for automatic linking of Employees to Managers, Departments, and Locations.

Step 3 – Prioritization

Selected Solution:

Automating Employee Data Onboarding using Import Sets and Dot Walking.

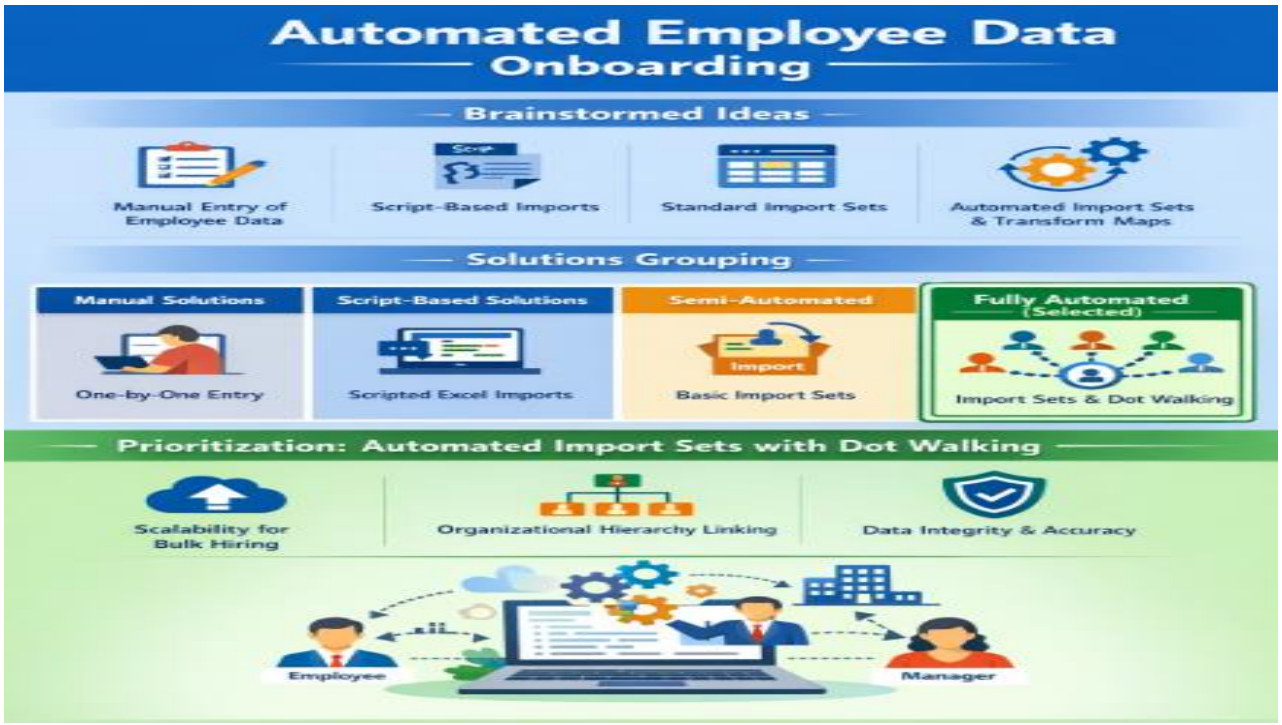
Reason:

Scalability: Handles bulk data during hiring seasons or company-wide updates easily.

Organizational Hierarchy: Dot Walking ensures that when an employee is imported, their relationship to their Manager and Department is automatically and accurately linked.

Data Integrity: Transform Maps prevent duplicate employee IDs and ensure proper formatting of email addresses and phone numbers.

Reduced Human Error: Minimizes the risk of sensitive employee data being incorrectly assigned.



2.2DEFINE PROBLEM STATEMENT

PROBLEM STATEMENT - 1 : HR Administrator / Data Manager
I am: An HR Administrator or Data Operations Manager.
I'm trying to: Maintain accurate and up-to-date employee records across multiple organizational platforms (like payroll, attendance, and internal directories).
But: Manually entering and updating employee data is repetitive and highly prone to human error.
Because: Data is currently scattered across various spreadsheets and manual forms, making real-time synchronization nearly impossible.
Which makes me feel: Overwhelmed by administrative overhead, and frustrated because the data is often inconsistent or outdated when needed for urgent reporting.

PROBLEM STATEMENT - 2 :

IT Support / Security Officer
I am: An IT Systems Administrator or Security Officer.
I'm trying to: Automatically grant or revoke access to company software (Slack, Zoom, GitHub, etc.) whenever a new employee joins or someone leaves.
But: Doing this manually for every single tool takes forever, and sometimes we forget to remove access for employees who have left the company.
Because: There is no "bridge" between the HR database and our IT software systems, so I have to create accounts one by one.
Which makes me feel: Stressed about security risks (data leaks) and annoyed by the constant "I can't log in" tickets from new joiners.

2.3EMPATHY MAP CANVAS

User: HR Administrator / Data Manager

Says:

- Updating employee data takes too much time.
- Generating reports is difficult and a struggle.
- The data is sometimes incorrect or inconsistent.

Thinks:

- It would be great to have a system that reduces manual work.
- Life would be much easier if data updated automatically.

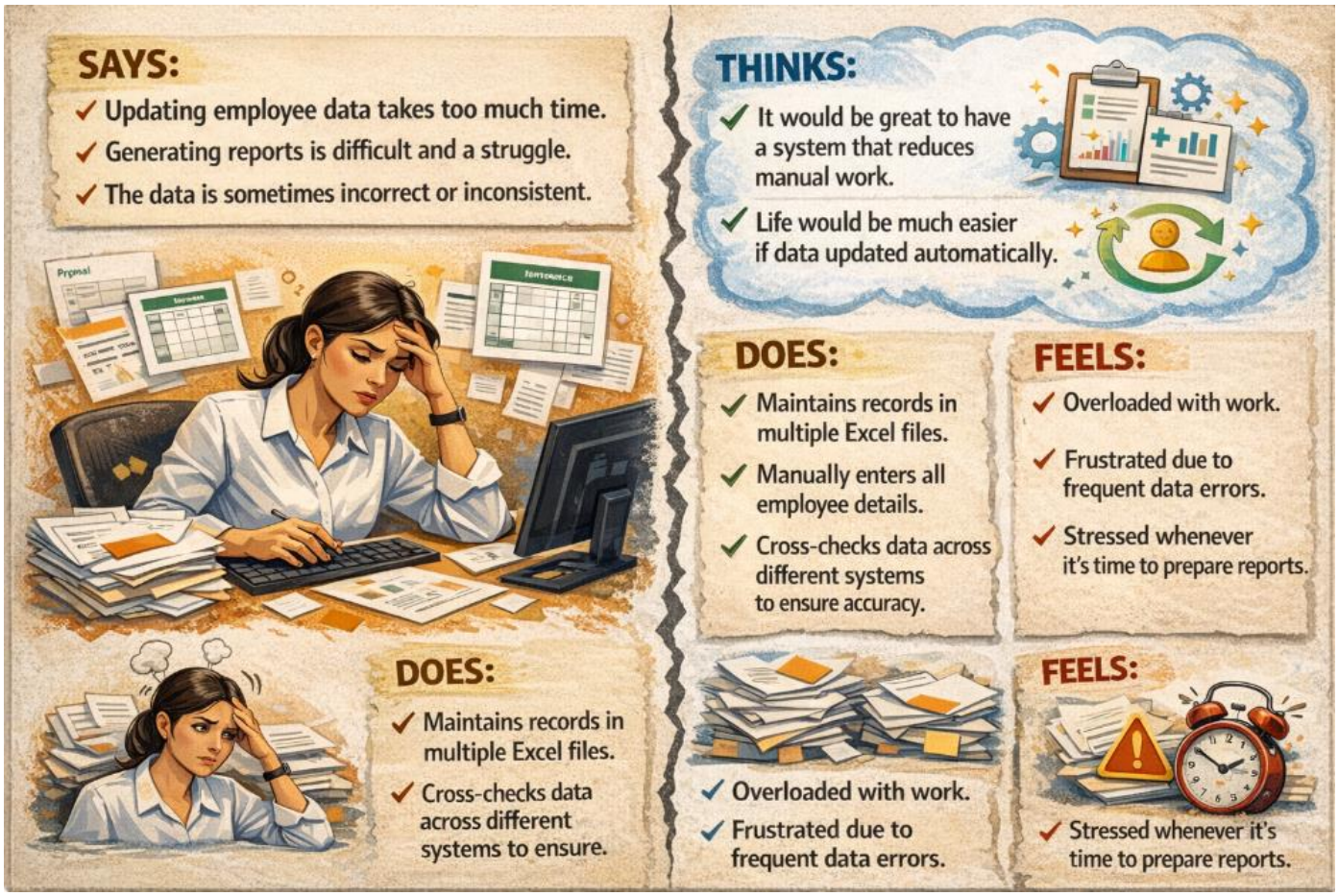
Does:

- Maintains records in multiple Excel files.
- Manually enters all employee details.
- Cross-checks data across different systems to ensure accuracy.

Feels:

- Overloaded with work.
- Frustrated due to frequent data errors.

Stressed whenever it’s time to prepare reports



3.REQUIREMENT ANALYSIS

3.1. .DATA FLOW DIAGRAM & USER STORIES

3.1.1 USER STORY

In the **Automating Employee Data** system, the HR Administrator can add, update, and delete employee details to ensure that all employee records remain accurate and up-to-date across different organizational systems. The HR team can also generate reports such as attendance, payroll, and employee summaries automatically, which reduces manual work, minimizes errors, and saves time.

The system can integrate with **Microsoft Excel, Google Sheets, HR management software like Zoho People or BambooHR, and ServiceNow** for seamless data management and workflow automation, making it easier to handle large volumes of employee information.

Employees have the ability to view their personal details, payslips, and leave balances at any time, and they can update their contact and emergency information directly in the system.

This self-service feature ensures that the HR database always contains correct and current information, reducing the need for back-and-forth communication with HR. Using **ServiceNow portals or mobile apps**, employees can access their information in real time, submit requests, and track the status of HR services efficiently.

The IT Administrator can automatically assign or revoke access to company software like **Slack, Zoom, GitHub, Microsoft Teams**, and internal tools using **ServiceNow workflows** whenever employees join or leave.

3.1.2 DATA FLOW DIAGRAM (DFD)

EXTERNAL ENTITY:

The **external entities** in the Automating Employee Data system include the HR Administrator, Employee, and IT Administrator. The HR Administrator is responsible for adding, updating, and managing employee records, as well as generating reports and monitoring data accuracy. Employees interact with the system to view their personal details, payslips, leave balances, and to update their own information when needed. The IT Administrator manages system access by assigning or revoking permissions for company software and ensures that security protocols are followed. Together, these entities interact with the system to maintain accurate, secure, and up-to-date employee data while supporting smooth HR and IT operations.

Process 1 – Employee Data Management:

This process handles all employee-related information in a centralized system. HR administrators can add new employees, update existing records, or delete outdated details, ensuring that all employee data remains accurate and up-to-date across the organization.

Process 2 – Payroll & Attendance Integration:

This process automatically synchronizes employee attendance and payroll data. By integrating with attendance systems and payroll databases, the system ensures that salaries, leave balances, and reports are accurate, reducing manual calculations and errors.

Process 3 – Document Generation:

In this process, the system automatically generates essential documents such as payslips, offer letters, contracts, and performance reports. This eliminates manual document preparation, saves time, and ensures consistent formatting and accuracy.

Process 4 – Access Control Management:

ServiceNow workflows are used in this process to automatically

assign or revoke access to company software like Slack, Zoom, GitHub, or internal tools whenever employees join or leave. This improves security, reduces manual account management, and ensures timely access for new employees.

Process 5 – Notifications & Alerts:

This process sends notifications and alerts to HR, IT, and employees whenever critical updates occur, such as a new joiner, changes in employee data, or revoked access. It helps ensure timely action, improves communication, and reduces the risk of missed updates.

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      HR Admin / Employee enters data
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                      v
         [Employee Data Management]
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                      v
      Payroll & Attendance Integration
                      |
                      v
Document Generation (Payslip, Offer Letter, Reports)
                      |
                      v
    Access Control Management (ServiceNow)
                      |
                      v
  Notifications & Alerts to HR / IT / Employee
                      |
                      v
                    End
```

3.2 SOLUTION REQUIREMENTS

3.2.1 Functional Requirements

1. **Employee Registration**
   - The system should allow adding new employee details such as name, employee ID, department, designation, salary, and contact information.

2. **Employee Data Management**
   - The system should allow users to **view, update, and delete employee records** when necessary.

3. **Automated Data Update**
   - The system should automatically update employee information such as salary changes, department transfers, and promotions.

4. **Attendance Management**
   - The system should record and store employee attendance details.

5. **Payroll Processing**
   - The system should automatically calculate employee salaries based on attendance, allowances, and deductions.

6. **Report Generation**
    - The system should generate reports such as:
    - Employee list report
    - Salary report
    - Attendance report

7. **Search Functionality**
    - Users should be able to search employee details using employee ID, name, or department.

8. **User Authentication**
    - The system should allow only authorized users (admin/HR) to access and manage employee data.

9. **Data Validation**
    - The system should check and validate employee information to avoid incorrect or duplicate data.

10.Data Storage

- The system should store employee information securely in a database.

3.2.2 Non-Functional Requirements

Non-functional requirements describe **how the system performs its functions**.

1. **Performance**
   - The system should process employee data quickly and generate reports without delay.

2. **Security**
   - Only authorized users (Admin/HR) should be able to access or modify employee data.

3. **Reliability**
   - The system should store employee data safely and prevent data loss.

4. **Usability**
   - The system interface should be simple and easy to use for HR staff.

5. **Scalability**
   - The system should support a large number of employees as the organization grows.

6. **Data Backup**
    - The system should regularly back up employee records to avoid data loss.

3.3 TECHNOLOGY STACK

3.3.1 ServiceNow Platform

The primary platform utilized for this project is ServiceNow, a leading cloud-based Enterprise Service Management (ESM) tool. It provides a robust framework for automating complex business workflows and centralizing data management. In this project, ServiceNow acts as the core engine to streamline employee data handling, ensuring data integrity and reducing manual intervention.

3.3.2 Automating Employee Data Management

To achieve seamless automation of employee records, the following ServiceNow components are implemented:

Import Sets:

These serve as a staging area to retrieve data from external sources (like Excel, CSV, or XML). It ensures that raw data is verified before being committed to the permanent database.

Transform Maps:

A critical automation component that maps the source data fields to the target User [sys_user] table. It uses logic to determine whether to create new records or update existing ones (Coalesce).

Scheduled Imports:

To make the process fully automated, scheduled jobs are configured to pull employee data at specific intervals, ensuring the system always reflects the most current information.

3.3.3 Business Benefits

Efficiency:

Eliminates the need for manual data entry, saving significant administrative time.

Accuracy:

Reduces human errors during data migration and updates.

Centralization:

Provides a "Single Source of Truth" for all employee-related information within the organization.

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Problem Statement:

Organizations frequently struggle with the manual synchronization of employee data from external HR or legacy systems into ServiceNow. Manual entry is not only time-consuming but highly prone to human error, resulting in duplicate records, broken data references, and a lack of real-time visibility into employee organizational hierarchies.

The Solution:

This project implements an automated pipeline using Import Sets and Transform Maps. By defining specific staging tables and transformation logic, we ensure that data is cleansed and validated before hitting the production tables. Furthermore, by utilizing Dot Walking, we enable the system to dynamically pull related data (like Department or Manager details) directly onto Incident or Change forms, eliminating the need for users to manually search for reference records.

4.2 PROPOSED SOLUTION

Parameter	Description
Problem	Manual entry of employee data causing errors and duplicates.
Solution	Automated import using ServiceNow Import Sets.
Technology	ServiceNow Platform and Transform Maps.
Key Feature	Dot Walking for real-time relationship mapping.
Benefit	Faster access to accurate employee details and records
Scalability	Capable of handling large and growing employee datasets

4.3 **Solution Architecture**

The architecture consists of multiple layers that automate the data import and relationship mapping process:

External Data Layer:

Employee data stored in spreadsheet files (Excel/CSV).

Import Layer:

Import Sets capture the file and store it in a temporary staging table.

Transformation Layer:

Transform Maps convert staging data into ServiceNow target records.

Database Layer:

Employee records are finalized and stored in the User (sys_user) table.

Application Layer:

Forms retrieve and display related manager/department info using Dot Walking.

Configuration Layer:

Dictionary Overrides customize field behaviors for extended tables

5. PROJECT PLANNING PHASE

5.1 CREATE RECORDS IN THE SHEETS

1. Open google sheet

2. Name the fields as name,email,department manager and manager's email

3. Enter some records

4. Save the sheet as: employee data

5. Download the file as .xlsx , .csv (or) .xml file

5.2 IMPORT DATA

1. Open service now.

2. Click on All >> search for import sets

3. Select load data under system import sets

4. Select create table

5. Give the the import set table name as: import employee

6. Choose the file which we created in the XL sheet.

7. The data is imported,click on personalized list

8. Add fields to the form.

5.3 CREATE TRANSFORM MAP

1. Open service now.

2. Click on All >> search for import set

3. Select create transform map under import sets

4. Click on new

5. Fill the following details to create a new transform map

6. Give name as: Employee data transform

7. Select source table: import employee data

8. Select target table: user[sys_user]

9. Save the form.

5.4 MATCH THE FIELDS USING MAPPING ASSIST

1. Scrolled down the created transform map form

2. Click on auto matching fields

3. Then in the mapping fields some fields are matched

4. Using mapping assist map the manager field.

5. Select the fields which you are mapping in the staging table and target table

6. Select manager in source table and add

7. Select manager field in target table

8. Click on the save button.

5.5 TRANSFORM THE DATA

1. Click on transform in the related links.

2. Click on transform.

3. Data's transform state is completed.

4. Click on ISET0010003 under next steps

5. Two rows are inserted.

6. PROJECT DEVELOPMENT PHASE

6.1. PERFORMANCE TESTING

6.1.1 Project Development

1.Problem Statement

2.Empathy Map Canvas

3.Brainstroming and Idea Prioritization

4.Data Flow Diagram & User Stories

5.Solution Requirement

6.Technology Stacks

Why you'll learn

1.Advanced Features Integration

2.Mobile Application Development

3.Enhanced Security Protocols

4.User Interface Optimization

5.Cloud Storage Integration

6.Performance Optimization

7.Scalability & database Requirements

8.Custom Functional Requirements

9.System Reliability & Updates

10.Technological Expansion

7.ADVANTAGES AND DISADVANTAGES

Advantages

- Reduces manual work
- Improves data accuracy
- Faster processing of employee data
- Centralized and secure system
- Easy report generation

Disadvantages

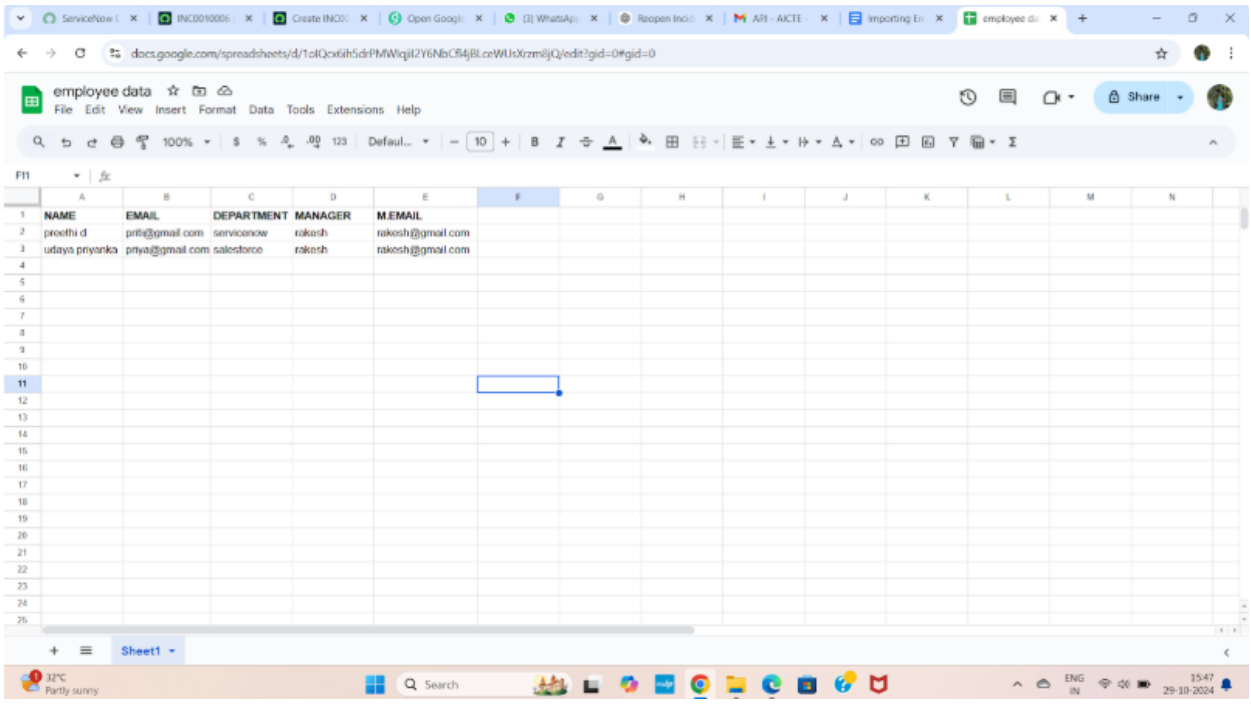
- Complex setup
- High cost
- Requires training
- Needs regular maintenance
- Data migration risks

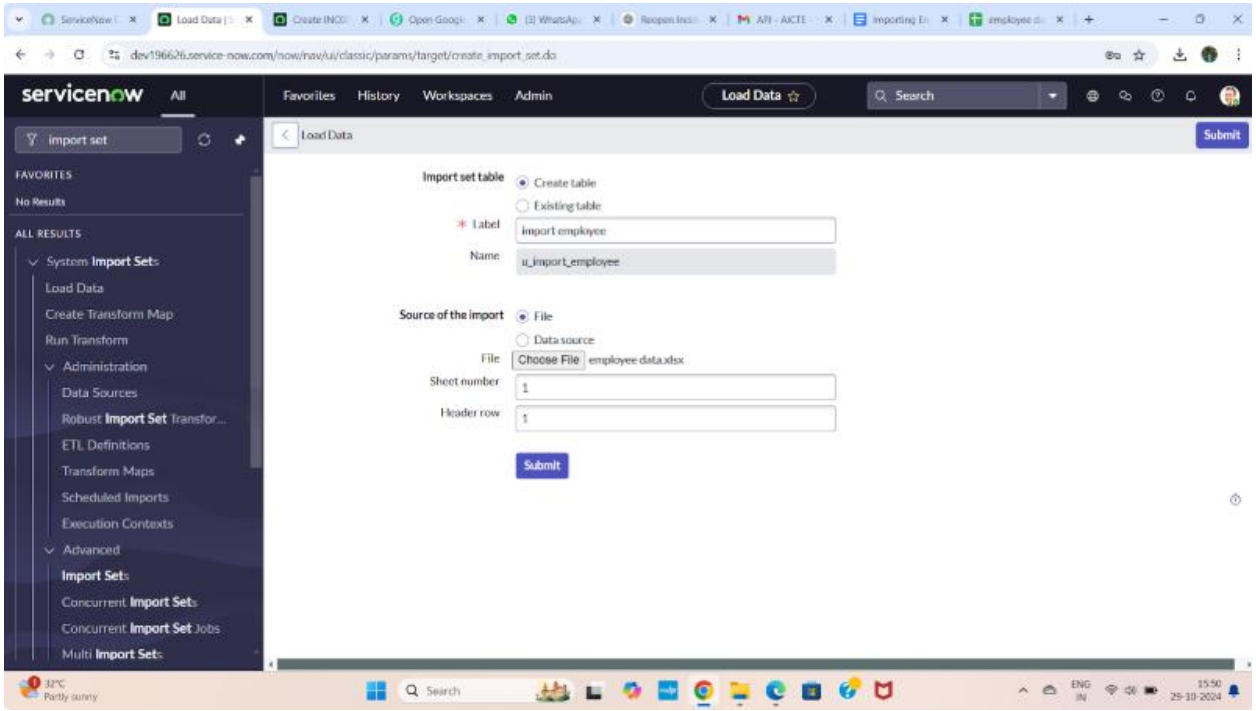
8.PROJECT DOCUMENTATION

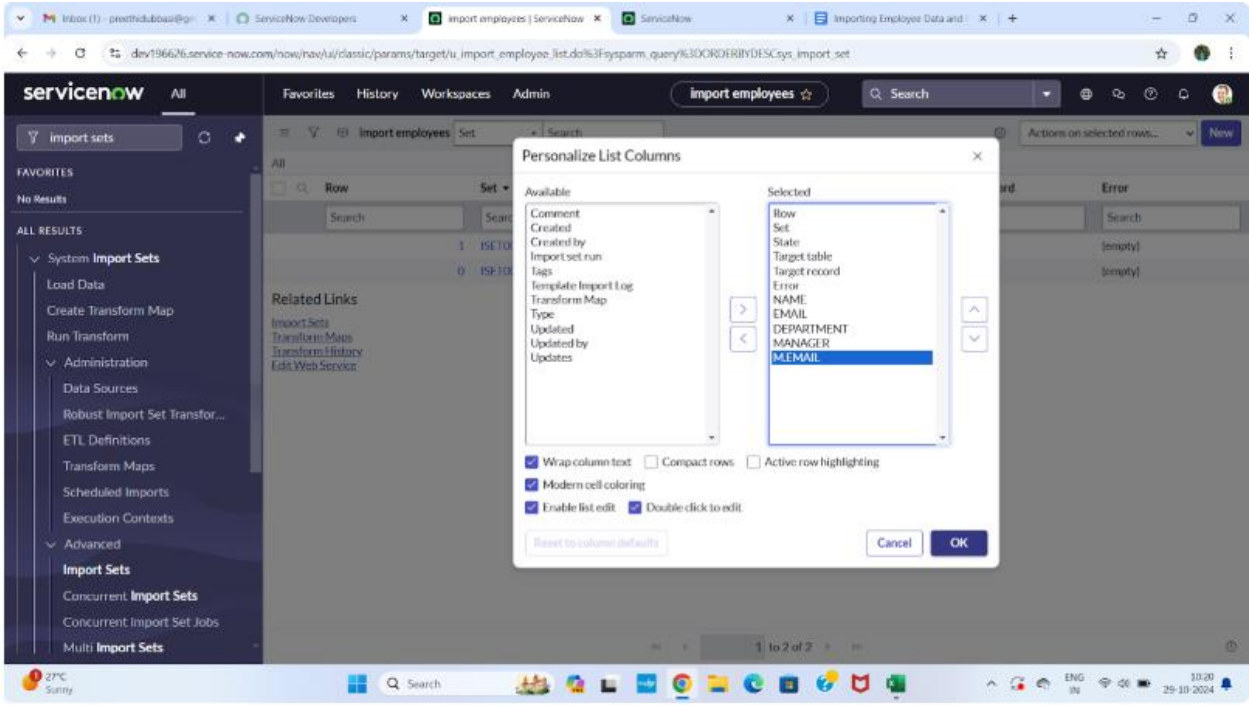
8.1 FINAL REPORT

The **Automating Employee Data project using ServiceNow** focuses on improving the efficiency of managing employee information by reducing manual work and errors. In traditional systems, employee data is handled manually, which often leads to data inconsistency, time consumption, and difficulty in generating reports. This project aims to automate employee data entry, storage, and processing to ensure accuracy, security, and faster access to information. The system is developed using ServiceNow with technologies such as Service Portal and UI Builder for the frontend, JavaScript and Business Rules for backend processing, and ServiceNow tables for database management. The methodology involves using Import Sets to upload bulk employee data, Transform Maps to clean and map the data correctly, and Dot Walking to automatically link employees with their managers and departments. The system includes modules like employee management, attendance management, payroll processing, report generation, and user authentication. It offers several advantages such as reduced manual effort, improved data accuracy, faster processing, and secure data handling. However, it also has some limitations, including complex initial setup, higher implementation cost, and the need for user training. Overall, this project successfully demonstrates how automation in ServiceNow can streamline employee data management and enhance organizational productivity.

8.2 SCREEN LAYOUT



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Import set run
Tags
Template Import Log
Transform Map
Type
Updated
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Row
Set
State
Target table
Target record
Error
NAME
EMAIL
DEPARTMENT
MANAGER
M.EMAIL
Wrap column text
Compact rows
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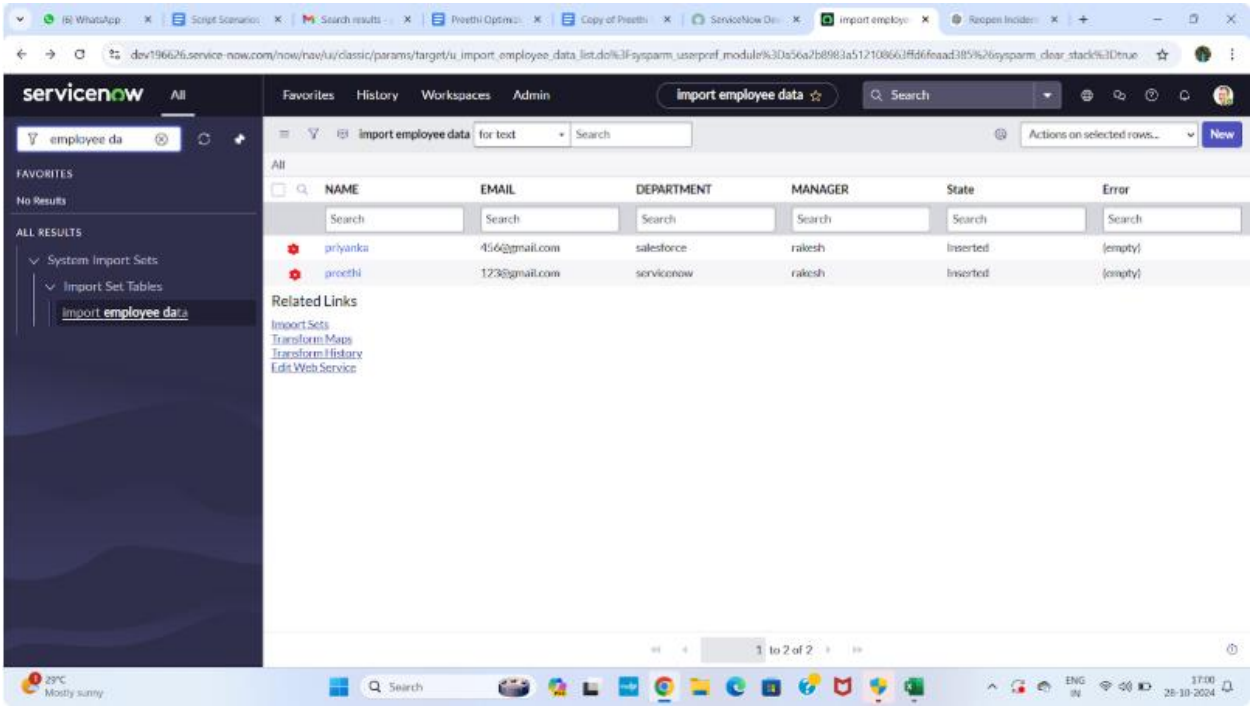
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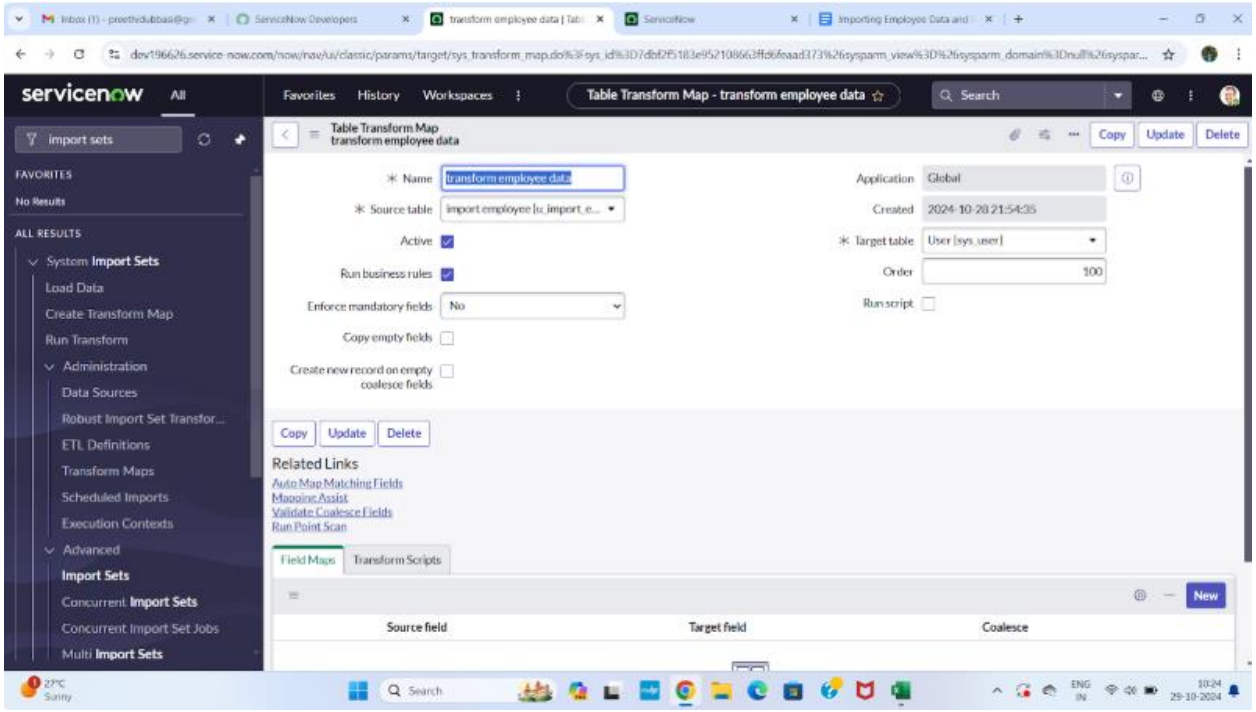
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Table Transform Map
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Copy
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Source table
import employee [u_import_e...
Active
Run business rules
Enforce mandatory fields
No
Copy empty fields
Create new record on empty coalesce fields
Application
Global
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Target table
User [sys_user]
Order
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Auto Map Matching Fields
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Run Point Scan
Field Maps
Transform Scripts
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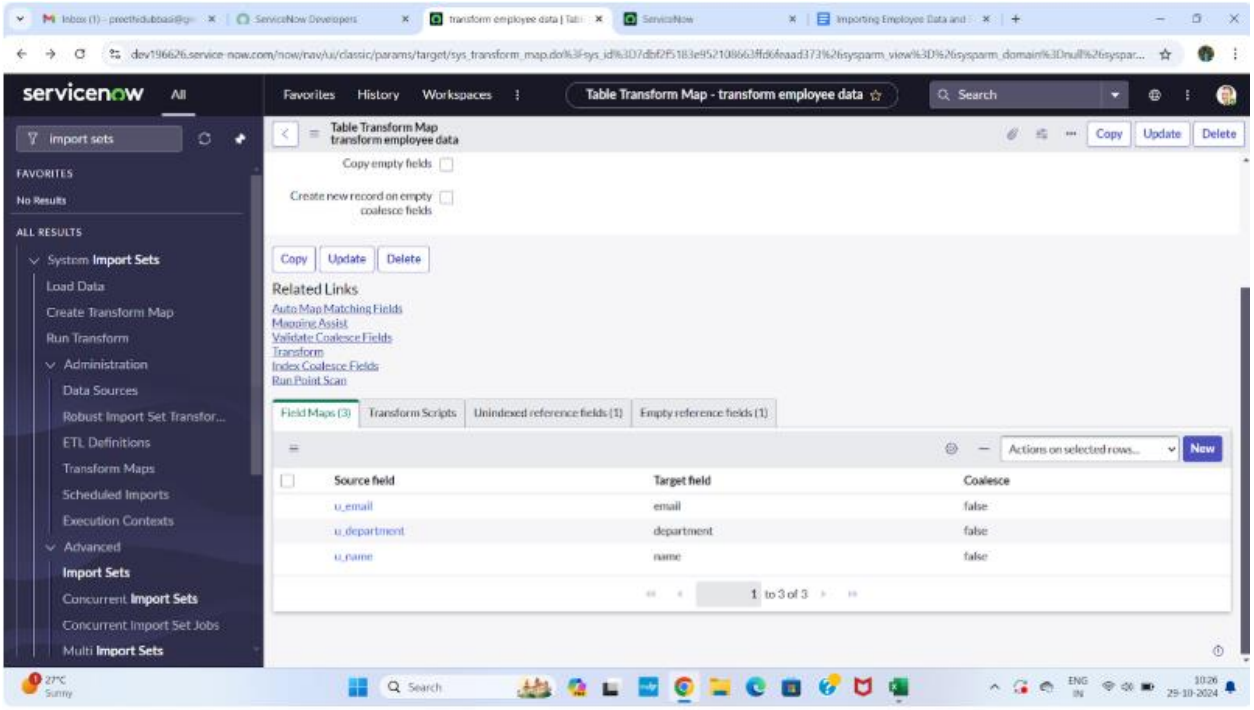
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Table Transform Map
transform employee data
Copy
Update
Delete
Copy empty fields
Create new record on empty coalesce fields
Related Links
Auto Map Matching Fields
Mapping Assist
Validate Coalesce Fields
Transform
Index Coalesce Fields
Run Point Scan
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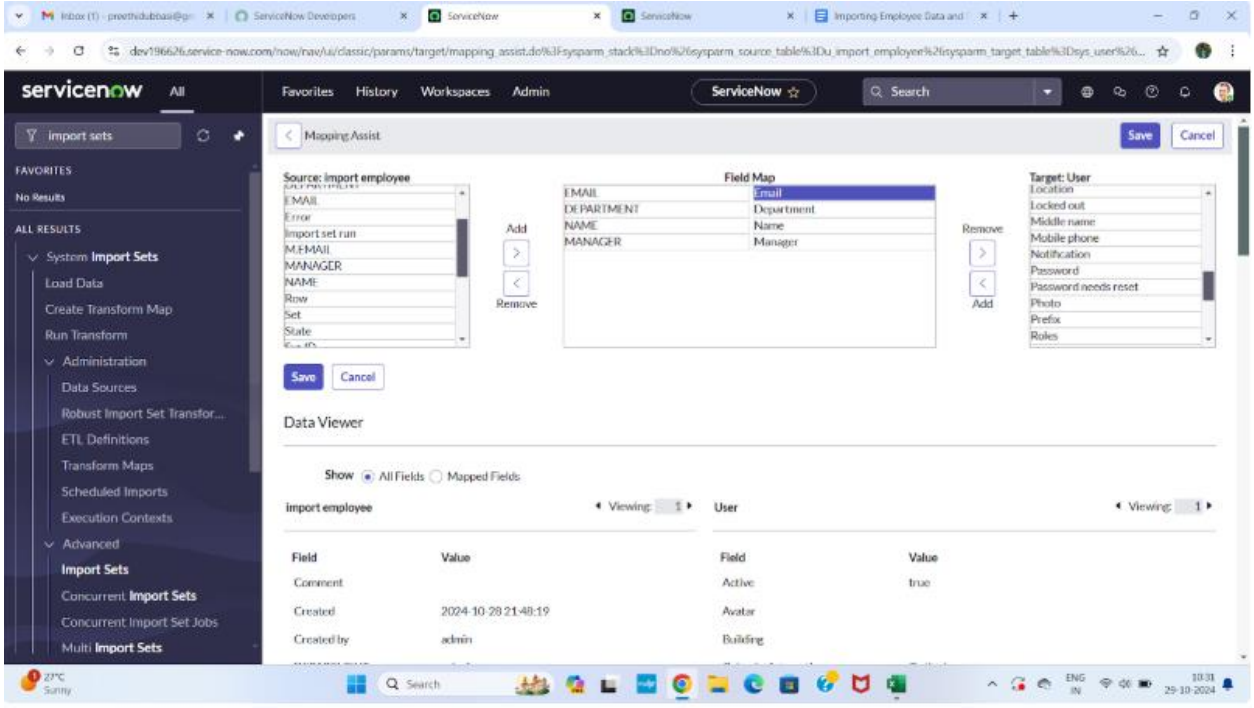
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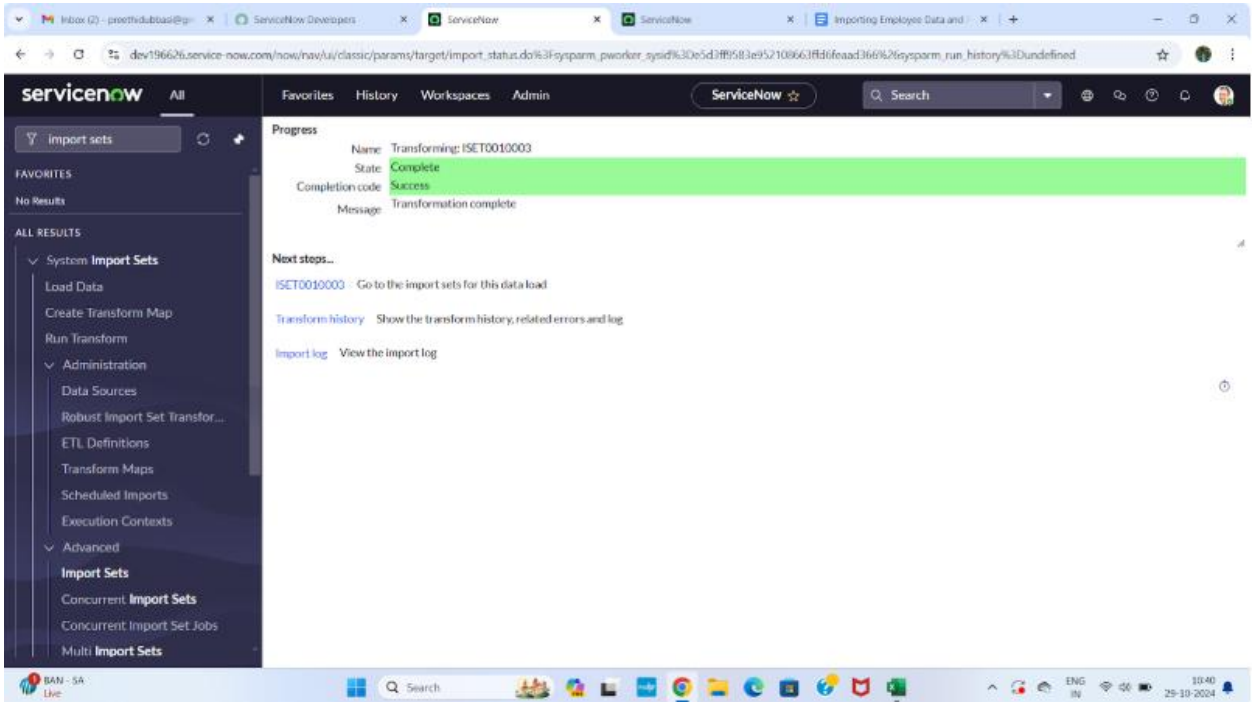
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ETL Definitions
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Multi Import Sets
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Message
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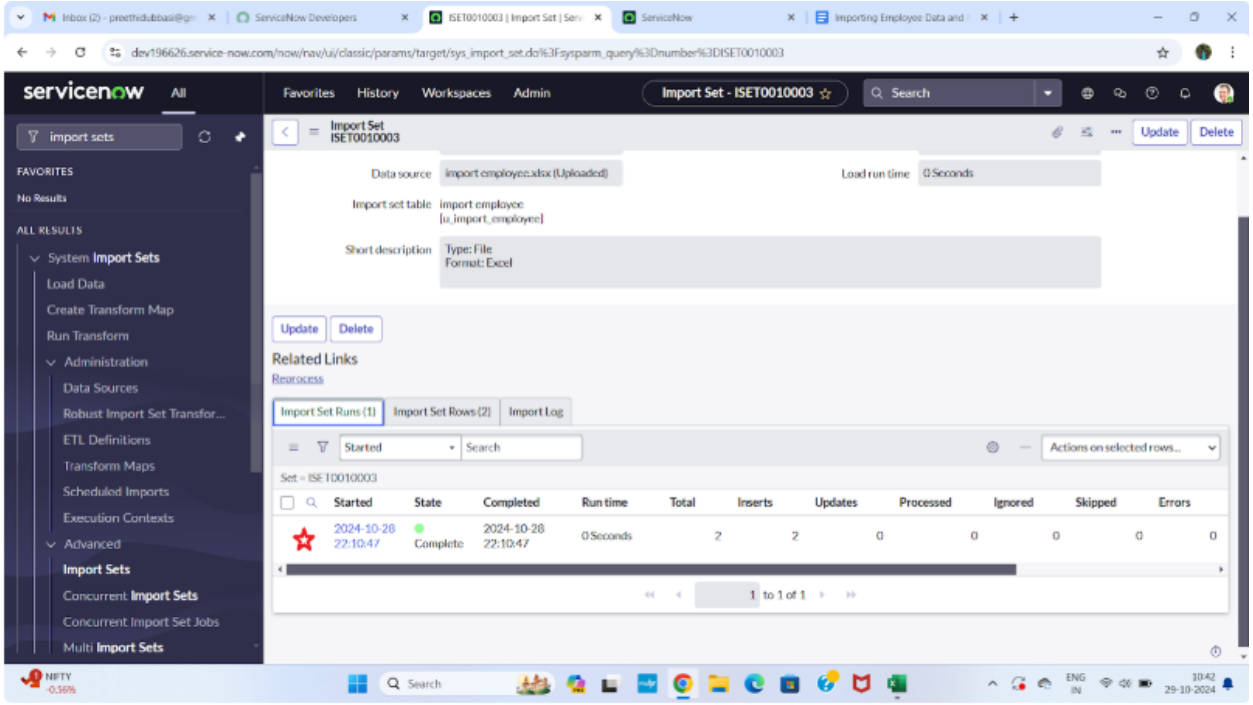
Go to the import sets for this data load
Transform history
Show the transform history, related errors and log
Import log
View the import log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Transform Maps
Scheduled Imports
Execution Contexts
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Import Sets
Concurrent Import Sets
Concurrent Import Set Jobs
Multi Import Sets
Import Set
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Delete
Data source
import employee.xlsx (Uploaded)
Load run time
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Import set table
import employee
[u_import_employee]
Short description
Type: File
Format: Excel
Related Links
Reprocess
Import Set Runs (1)
Import Set Rows (2)
Import Log
Started
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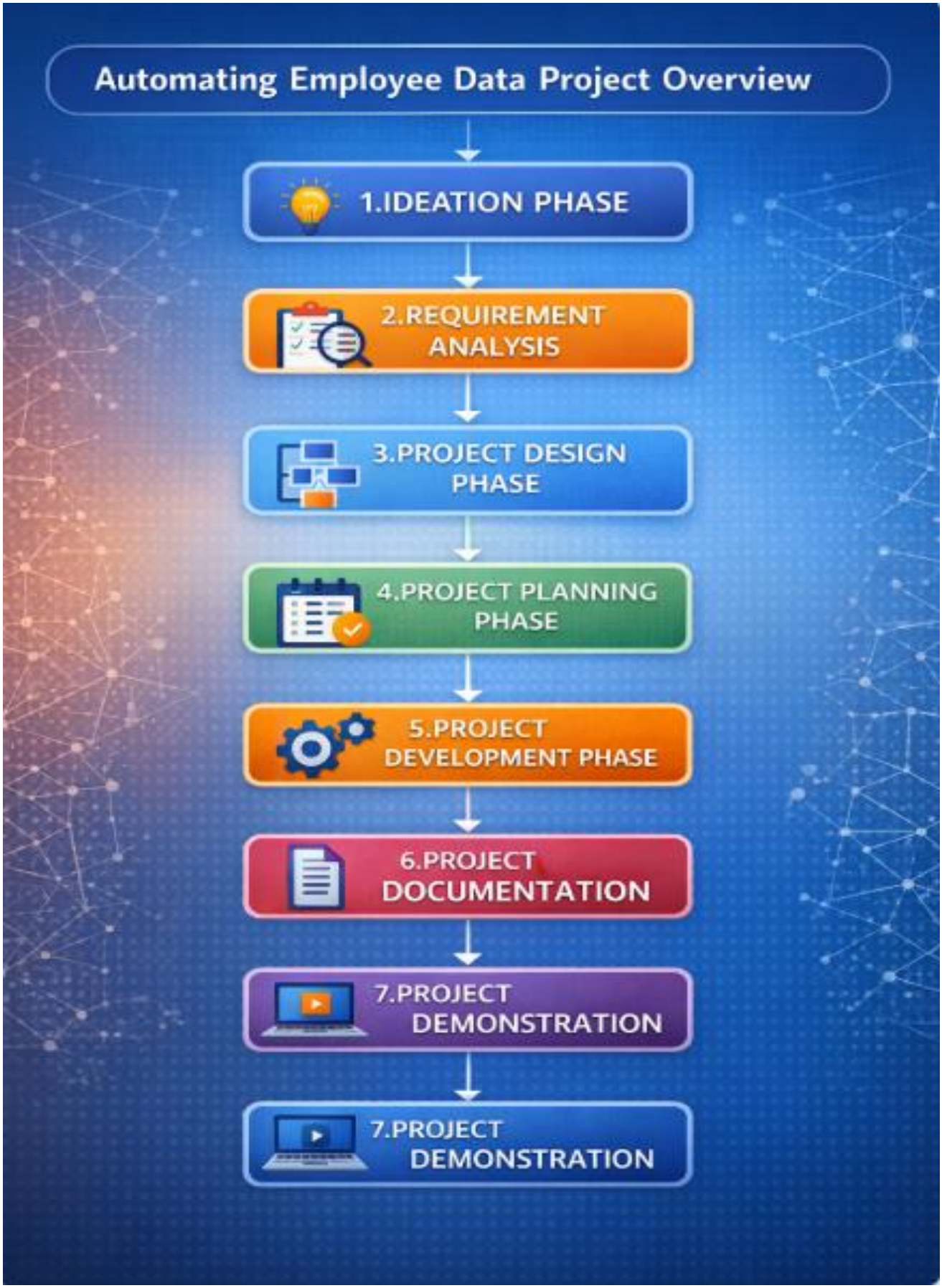
9. CONCLUSION

The project "Automating Employee Data" was successfully implemented to simplify and streamline the process of managing employee information. The main objective of this project was to eliminate manual data entry and automate the storage, updating, and retrieval of employee records.

By using automation techniques, the system ensures accurate, consistent, and efficient handling of employee data across different departments. It reduces human errors, saves time, and improves overall productivity. The system also maintains proper relationships between employee details such as roles, departments, and records.

This project demonstrates how automation can enhance data management in modern organizations. It highlights the importance of adopting digital solutions to handle large volumes of data efficiently, ensuring reliability, scalability, and better performance in organizational processes.


Automating Employee Data Project Overview
1.IDEATION PHASE
2.REQUIREMENT ANALYSIS
3.PROJECT DESIGN PHASE
4.PROJECT PLANNING PHASE
5.PROJECT DEVELOPMENT PHASE
6.PROJECT DOCUMENTATION
7.PROJECT DEMONSTRATION
7.PROJECT DEMONSTRATION